

ROADWATCH – SEMESTER SPRINT / ITERATION PLAN

Project: RoadWatch – Pothole Detection Using Mobile Dashcam Imagery with a Web-Based Monitoring System | **Methodology:** Agile / Scrum

Sprint	Weeks	Main Tasks	Expected Output
Sprint 1	1–2	Define requirements and system scope; research and prepare pothole dataset; finalize architecture.	Approved requirements, scope, and initial system design.
Sprint 2	3–4	Prepare/organize dataset; configure YOLO; begin model training and validation.	Initial pothole detection model and training results.
Sprint 3	5–6	Test model; convert to ONNX; integrate on-device inference into Flutter.	Working on-device pothole detection.
Sprint 4	7–8	Implement camera workflow, GPS, pothole alerts, detection saving, and basic mobile UI.	Functional mobile detection and location workflow.
Sprint 5	9–10	Develop FastAPI endpoints; configure PostgreSQL; implement detection storage and validation.	Working backend API and database.
Sprint 6	11–12	Connect Flutter to API; implement online/offline handling and synchronization.	Mobile-to-backend integration with detection syncing.
Sprint 7	13–14	Develop web monitoring dashboard; display saved pothole detections and locations.	Functional web-based monitoring dashboard.
Sprint 8	15–16	End-to-end integration testing; fix bugs; finalize documentation and presentation.	Tested final system, documentation, and presentation materials.

Iteration approach: Each sprint produces a usable increment that is reviewed, tested, and refined before the next sprint. Tasks may be adjusted based on testing results and project progress while keeping the final system scope focused on pothole detection.